

Math 241, F04
Quiz 4

Name:
SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (10 points) Given the position vector

$$\mathbf{r}(t) = t \cos t \mathbf{i} + t \sin t \mathbf{j} + \sqrt{2}t \mathbf{k}$$

a) Find the velocity $\mathbf{v}(t)$, speed $|\mathbf{v}(t)|$ and acceleration $\mathbf{a}(t)$.

Solution:

$$\begin{aligned}\mathbf{v}(t) &= \mathbf{r}'(t) = (\cos t - t \sin t) \mathbf{i} + (\sin t + t \cos t) \mathbf{j} + \sqrt{2} \mathbf{k} \\ |\mathbf{v}(t)| &= \sqrt{(\cos t - t \sin t)^2 + (\sin t + t \cos t)^2 + (\sqrt{2})^2} \\ &= \sqrt{(\cos^2 t - 2t \cos t \sin t + t^2 \sin^2 t) + (\sin^2 t + 2t \sin t \cos t + t^2 \cos^2 t) + 2} \\ &= \sqrt{t^2 \cos^2 t + t^2 \sin^2 t + \cos^2 t + \sin^2 t + 2} \\ &= \sqrt{t^2 + 3} \\ \mathbf{a}(t) &= \mathbf{r}''(t) = (-\sin t - \sin t - t \cos t) \mathbf{i} + (\cos t + \cos t - t \sin t) \mathbf{j} \\ &= (-2 \sin t - t \cos t) \mathbf{i} + (2 \cos t - t \sin t) \mathbf{j}\end{aligned}$$

b) Find the length of the curve between $0 \leq t \leq 2$. **Solution:**

$$\begin{aligned}S &= \int_0^2 |\mathbf{r}'(t)| dt \\ &= \int_0^2 \sqrt{t^2 + 3} dt \\ &= \left(\frac{t}{2} \sqrt{t^2 + 3} + \frac{3}{2} \ln |t + \sqrt{t^2 + 3}| \right) \Big|_{t=0}^{t=2} \\ &= \left(\sqrt{7} + \frac{3}{2} \ln(2 + \sqrt{7}) \right) - \frac{3}{2} \ln \sqrt{3} \\ &\approx 4.126\end{aligned}$$

Problem 2. (10 points)

a) Change $(4, 4\pi/3, -8)$ from cylindrical to Cartesian coordinates.

Solution: Since $r = 4$, $\theta = 4\pi/3$ and $z = -8$, we have

$$x = r \cos \theta = 4 \cos \frac{4\pi}{3} = 4\left(-\frac{1}{2}\right) = -2$$

$$y = r \sin \theta = 4 \sin \frac{4\pi}{3} = 4\left(-\frac{\sqrt{3}}{2}\right) = -2\sqrt{3}$$

$$z = z = -8$$

Then Cartesian coordinates are

$$(-2, -2\sqrt{3}, -8)$$

b) Change the Cartesian equation $x^2 + y^2 + 4z^2 = 10$ to spherical coordinate equation.

Solution: We know $x = \rho \sin \phi \cos \theta$, $y = \rho \sin \phi \sin \theta$ and $z = \rho \cos \phi$, then we have

$$\rho^2 \sin^2 \phi \cos^2 \theta + \rho^2 \sin^2 \phi \sin^2 \theta + 4\rho^2 \cos^2 \phi = 10$$

$$\rho^2 \sin^2 \phi + 4\rho^2 \cos^2 \phi = 10$$

$$\rho^2 + 3\rho^2 \cos^2 \phi = 10$$